

Laurent Magri-Stella

Ph.D. student in theoretical and numerical cosmology

Education

- 2024–2025 **Second Year of "Cosmos, Fields, Particles" Master**, University of Montpellier,
Courses in :
○ Cosmology, Quantum Field Theory, High Energy Astrophysics, Dark Matter, Advanced Particle Physics, Interstellar Medium, Stellar Atmosphere and Radiative Transfer
- 2023–2024 **Second Year of Physics Master's**, *Faculté des Sciences*, University of Aix-Marseille,
Courses in :
○ General Relativity, Relativistic Universe, Quantum Field Theory, Astroparticles and Primordial Cosmology, Advanced Quantum Mechanics, Gauge Theory and Standard Model
○ Doctoral courses : Observational Cosmology, Field Theoretical Aspects of General Relativity
○ Additional courses I followed : Introduction to physics beyond the Standard Model, Advanced Quantum Field Theory
- 2022–2023 **First Year of Physics Master's**, *Faculté des Sciences*, University of Aix-Marseille,
Optional courses chosen :
Special Relativity, Cosmology, Statistics and Data Analysis, Advanced Mathematical Methods
- 2020–2022 **Physics Bachelor**, *Faculté des Sciences*, University of Aix-Marseille, *with high honors*
Modelling option
- 2019–2020 **First Year René Descartes**, *Faculté des Sciences*, University of Aix-Marseille
Physics speciality

Individual and professional experiences

- March - **M2 Research Internship**, *Laboratoire d'Annecy de Physique des Particules*, LSST
August team, under the guidance of Narei LORENZO MARTINEZ and Vincent REVERDY
- 2025 ○ Perturbing the FLRW metric to 1st order, propagating the perturbation through the Christoffel symbols to the geodesic equations.
○ Defining a 3D grid containing values of the Newtonian potential ϕ and the scale factor $a(t)$ which will be used to propagate the photons thanks to a RK4 integrator.
○ Simulating the gravitational lensing effects to quantify the uncertainties generated by the hypothesis made in modern analysis, such as the halo mass functions and mass density profiles.

- March - July 2024 **M2 Research Internship**, *Laboratoire d'Astrophysique de Marseille*, équipe "Galaxies, Etoiles, Cosmologie", under the guidance of Pr. Raphael GAVAZZI
- Use of Horizon-AGN cosmological and hydrodynamical simulation images and generated deflection maps to numerically compute different orders of decomposition of the Taylor expansion of the transformation jacobian matrix (Convergence κ , Shear γ_1 and γ_2 , and the four flexion terms : 1-flexion: F_1, F_2 and 3-flexion G_1, G_2)
 - Generated maps of the distribution of those lensing quantities and used them to compute Galaxy-Shear, Galaxy-Convergence and Galaxy-Flexion correlation functions
 - Interpreting the measurements as an additional constraint on the distribution of dark matter around the galaxies from the simulated images from which we know all the details of the dark matter distribution
 - Realising the same measurements on images from Euclid to add constraints on the distribution of dark matter
 - Periodic presentation of the results to the Euclid@LAM group and discussion with the associated professors and researchers
- February 2024 **Computing project**, *Université d'Aix-Marseille*, Centre de Physique des Particules de Marseille, under the guidance of Pr. Julian BAUTISTA
- Use of existing code to forecast errors on the measurements of growth rates in the Universe thanks to peculiar velocities
 - Adapting the C code to Python and using new simulated data to predict the errors on DESI/ZTF/... surveys.
 - Using specific libraries to optimize and speed up the code by paralleling operations (JaX/PyTorch)
- October 2023 **Autumn School on the Two Infinities**, *Université d'Aix-Marseille*, Centre de Physique des Particules de Marseille
- Courses presenting recent advancements and theories about :
 - Neutrino physics and the DUNE experiment
 - Introduction to Gravitational Waves
 - Precision tests of fundamental symmetries
 - Heavy Flavour Physics : from CP violation to Intensity frontiers
- October 2023 **IRN Terascale Workshop**, *Université d'Aix-Marseille*, Centre de Physique des Particules de Marseille
- Series of conferences on different topics such as :
 - Beyond Standard Model Physics
 - Studying of Higgs decays
 - Advancements in Cosmology and Dark Energy
 - Techniques and methods to analyse various data sets
- July 2023 **Summer School on Gravitational Waves**, *Université Côte d'Azur*, organized by MaNiTou
- Theoretical courses on gravitational waves, deriving results from calculations in the General Relativity framework.
 - Study and analysis of data from gravitational waves sources given by intervening researchers.
- April - June 2023 **M1 Research Internship**, *Centre de Physique Théorique*, Cosmology research team, under the guidance of Pr. Christian MARINONI
- Theoretical study of the propagation of a light beam in Lemaitre-Tolman-Bondi space-time models:
 - First introduction to the General Relativity framework to try and compute a formula for the luminosity distance for an off-center observer in LTB space-times.
 - Numerical study of the spherical distribution of the expansion rate fluctuations across the Universe:
 - Determining, with the use of multipolar expansions (up to hexadecupole), CosmicFlows4 catalog data and Python, anisotropies in the local expansion rate.
 - Use of Monte-Carlo tests of the models to determine statistical errors.

- October **Week at the "Observatoire de Haute Provence"**
- 2022
- Observations of various stellar objects and galaxies
 - Image processing of the latter objects
 - Quick courses given by researchers on image processing and astronomy.
- 2022 **Development of a model of the solar system**
- Study of the formation of the Kirkwood Gaps, thanks to the gravitational resonance of Jupiter
 - Use of a fourth order Leapfrog integration method, model made with python
 - Use of PyTorch to implement GPU-accelerated computations in the simulation.
- Sept. 2022 **Team Captain for Aix-Marseille's team, French Physicists Tournament, University**
- Feb. 2023 **of Aix-Marseille**
- Preparation of experiments and theoretical studies of different problems.
 - Presentation and defense of the discoveries in front of a jury made of researchers, and against other teams of students from the whole country.
- Sept. - Dec. **Physics tutor, University of Aix-Marseille**
- 2022
- Holding tutorial classes for students from the first year of undergraduate (for physics-related units)
- June 2022- **Employee at Carrefour**
- August
- Register and Home Delivery service
- 2023
- Team work, efficiency, taking care of the client and of the work space.
- Since 2019 **Tutor, Middle school, High school, Undergraduate**
- Use of pedagogical methods to support and help students
 - Taught students from different academic backgrounds :
 - First year of maths and computer science undergraduate
 - Finance bachelor
 - Physics bachelor
 - Medical school

Languages

French Read, written, and spoken fluently

English Read, written, and spoken fluently

German High-school level

Mother tongue

TOEIC score of 955

Diverse skills

Python	★ ★ ★ ★ ★	$\LaTeX$	★ ★ ★ ★ ★
Julia	★ ★ ★ ★ ★	Bayesian methods	★ ★ ★ ★ ★
Statistics	★ ★ ★ ★ ★	Mathematica	★ ★ ★ ★
C++	★ ★ ★ ★	ML	★ ★ ★
Fortran	★ ★ ★ ★	Java	★ ★ ★
Video editing	★ ★ ★	Office	★ ★ ★

Some hobbies

- Sports Judo black belt, practicing for more than 14 years. I also love occasional rock climbing, swimming, boxing and weightlifting.
- Chess ELO Rating of 1900, playing for more than 3 years, former Marseille-Echecs club member
- Music Currently learning how to play the guitar
- Tabletop Story-writing and world-building
RPGs
- Book Fantasy and Sci-Fi
writing